Learning-Accelerated ADMM for Stochastic Power System Scheduling with Numerous Scenarios

Ali Rajaei, Olayiwola Arowolo, Jochen L. Cremer

IEEE Transactions on Sustainable Energy, 2025

PowerTech Kiel 2025.

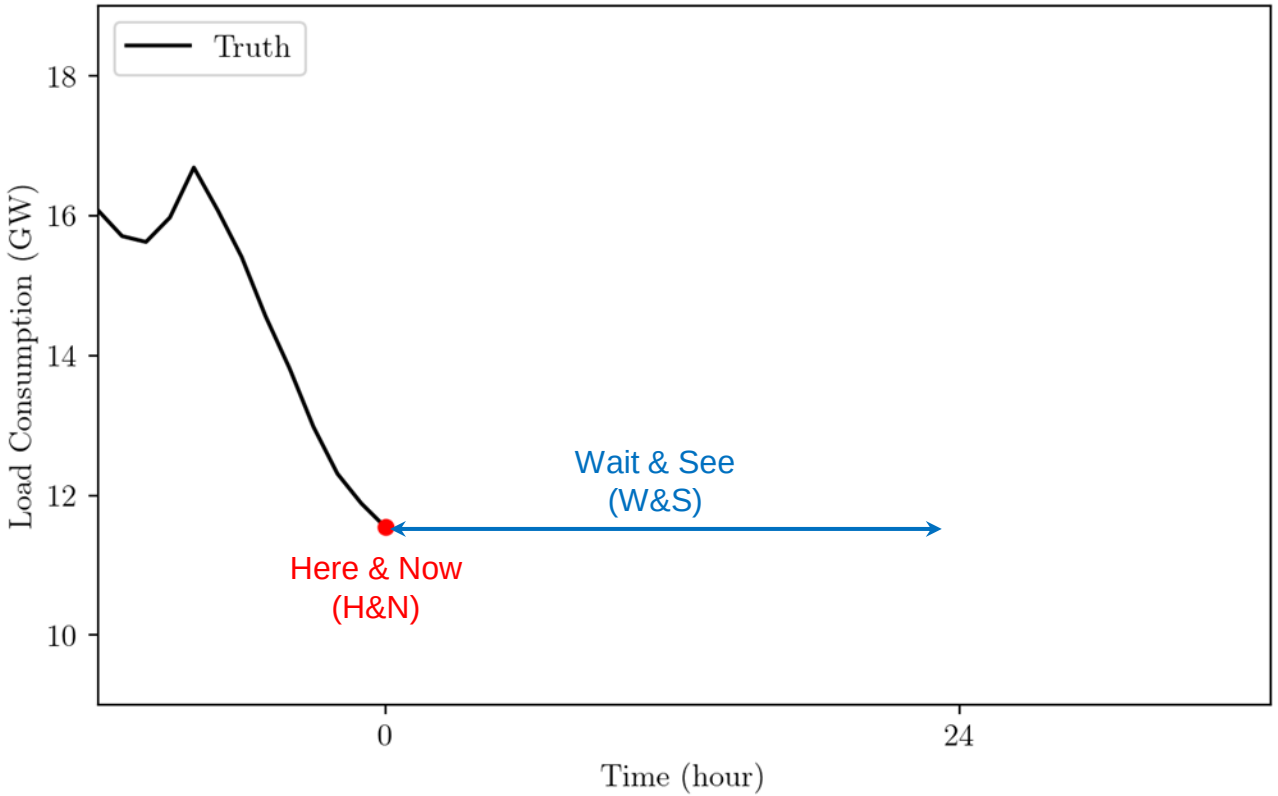
https://www.tudelft.nl/ai/delft-ai-energy-lab

Motivation

- RES integration and uncertainty

Motivation

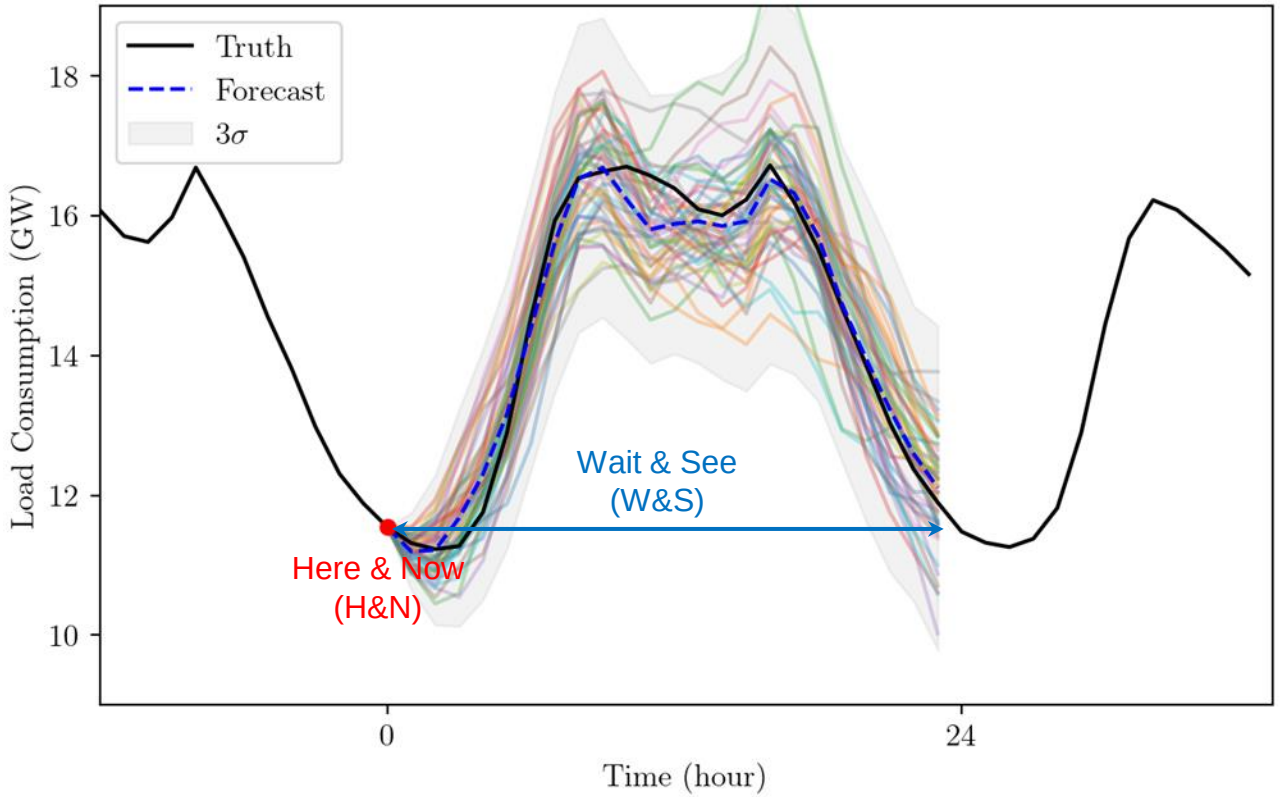
- RES integration and uncertainty
- Two-stage stochastic scheduling
  - Stochastic multi-period optimal power flow (St-MP-OPF)



TU Delft

Motivation

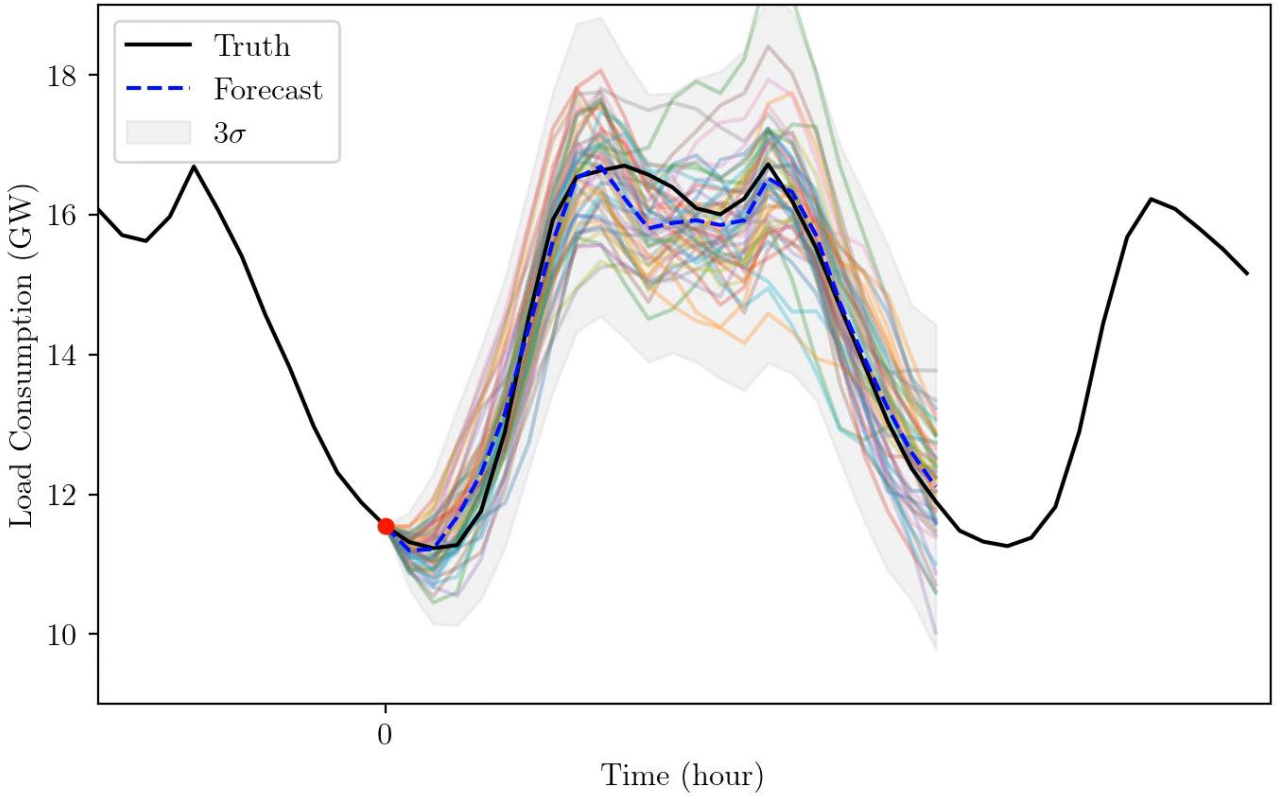
- RES integration and uncertainty
- Two-stage stochastic scheduling
  - Stochastic multi-period optimal power flow (St-MP-OPF)



TU Delft

Motivation

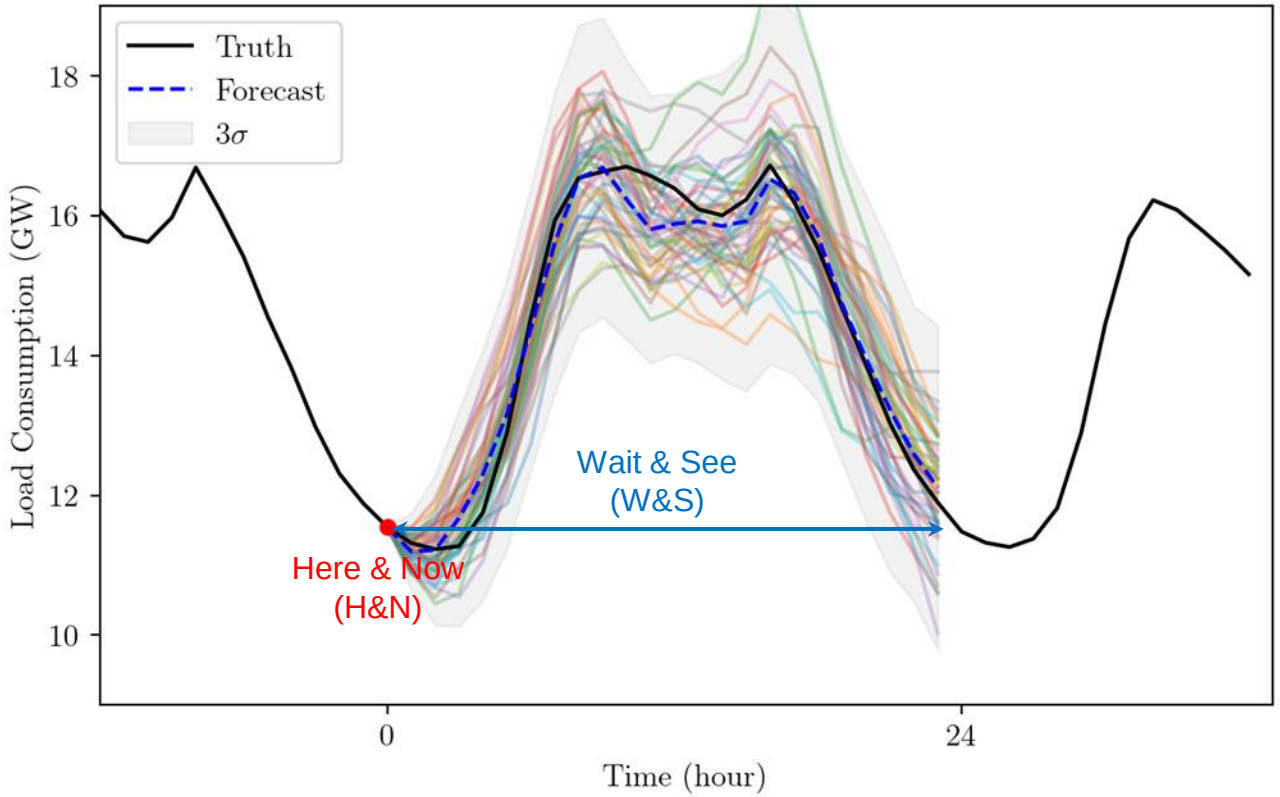
- RES integration and uncertainty
- Two-stage stochastic scheduling
  - Stochastic multi-period optimal power flow (St-MP-OPF)



TU Delft

Motivation

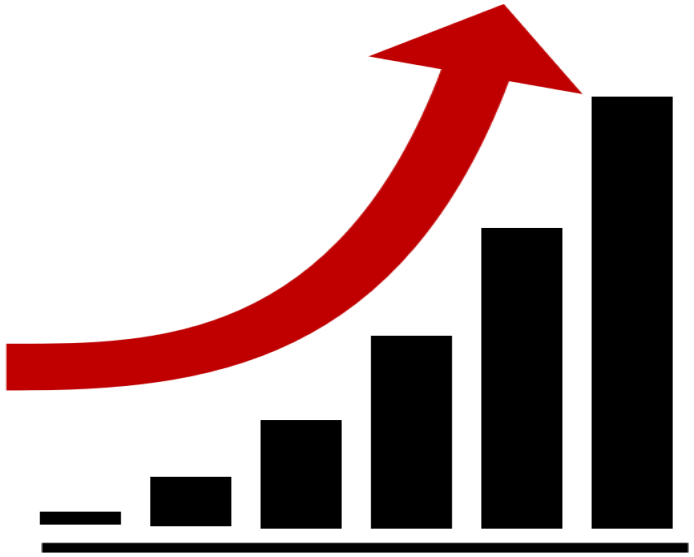
- RES integration and uncertainty
- Two-stage stochastic scheduling
  - Stochastic multi-period optimal power flow (St-MP-OPF)

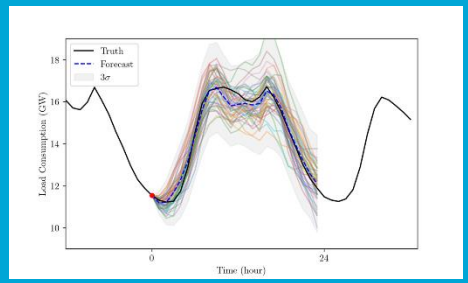


TU Delft

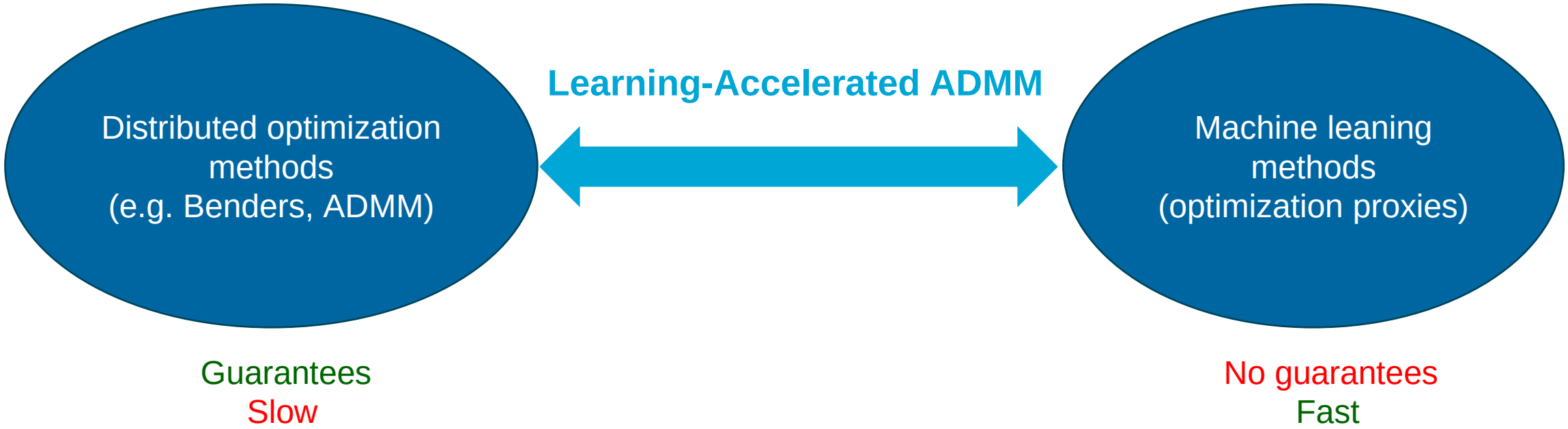
Motivation

- RES integration and uncertainty
- Two-stage stochastic scheduling
  - Stochastic multi-period optimal power flow (St-MP-OPF)
- Computational complexity in large-scale systems with numerous scenarios





TU Delft


Learning-Accelerated ADMM
Distributed optimization methods (e.g. Benders, ADMM)
Machine leaning methods (optimization proxies)
Guarantees
Slow
No guarantees
Fast

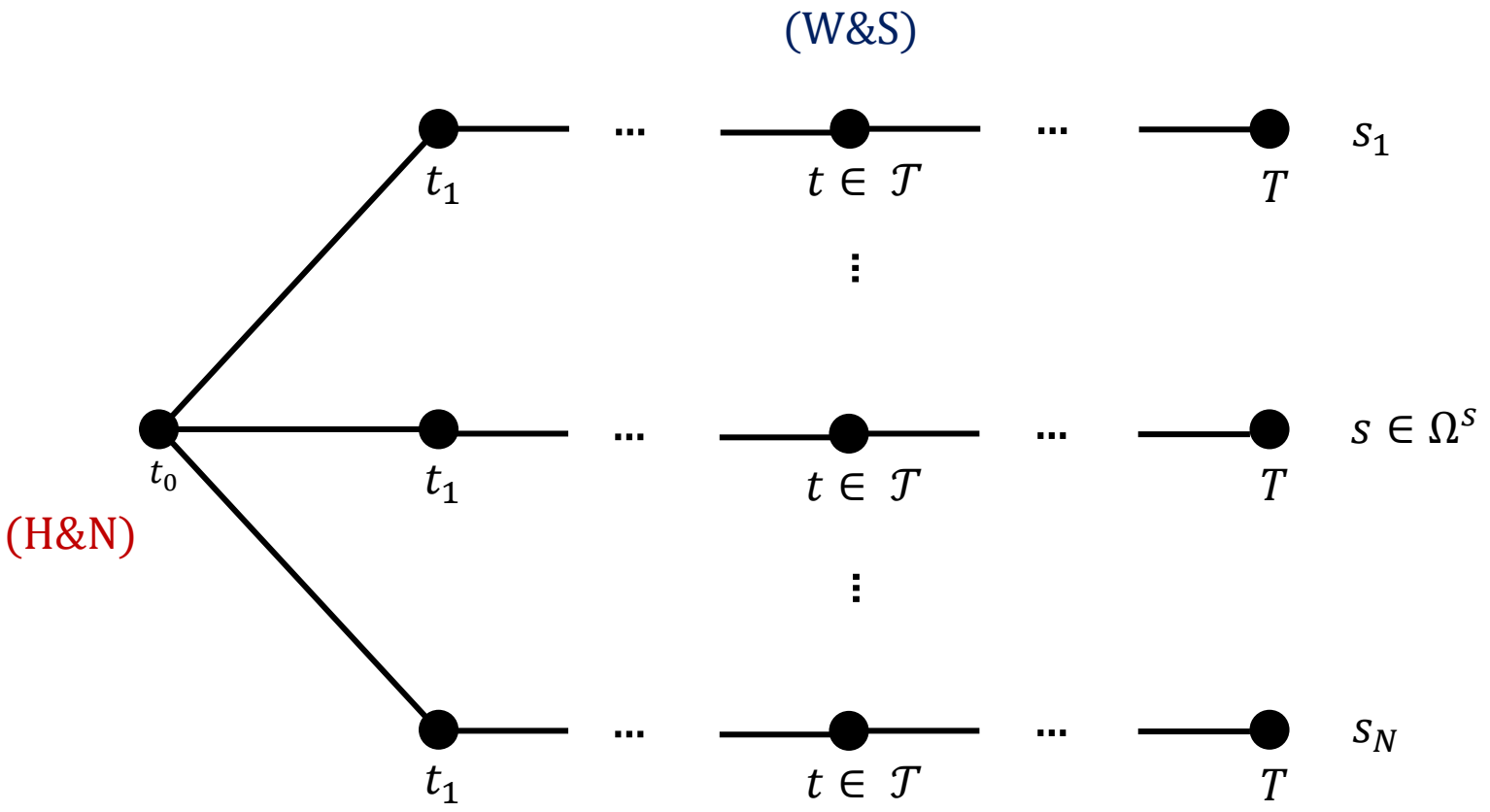
TU Delft

Stochastic scheduling

$$\min \sum_{s \in \Omega^s} \sum_{t \in \mathcal{T}} f(z_{t,s})$$

$$\text{s.t. } h(z_{t,s}) \leq 0 \qquad t \in \mathcal{T}, s \in \Omega^s$$

$$x = z_{t0,s} : \lambda_s \qquad t \in \mathcal{T}$$



TU Delft

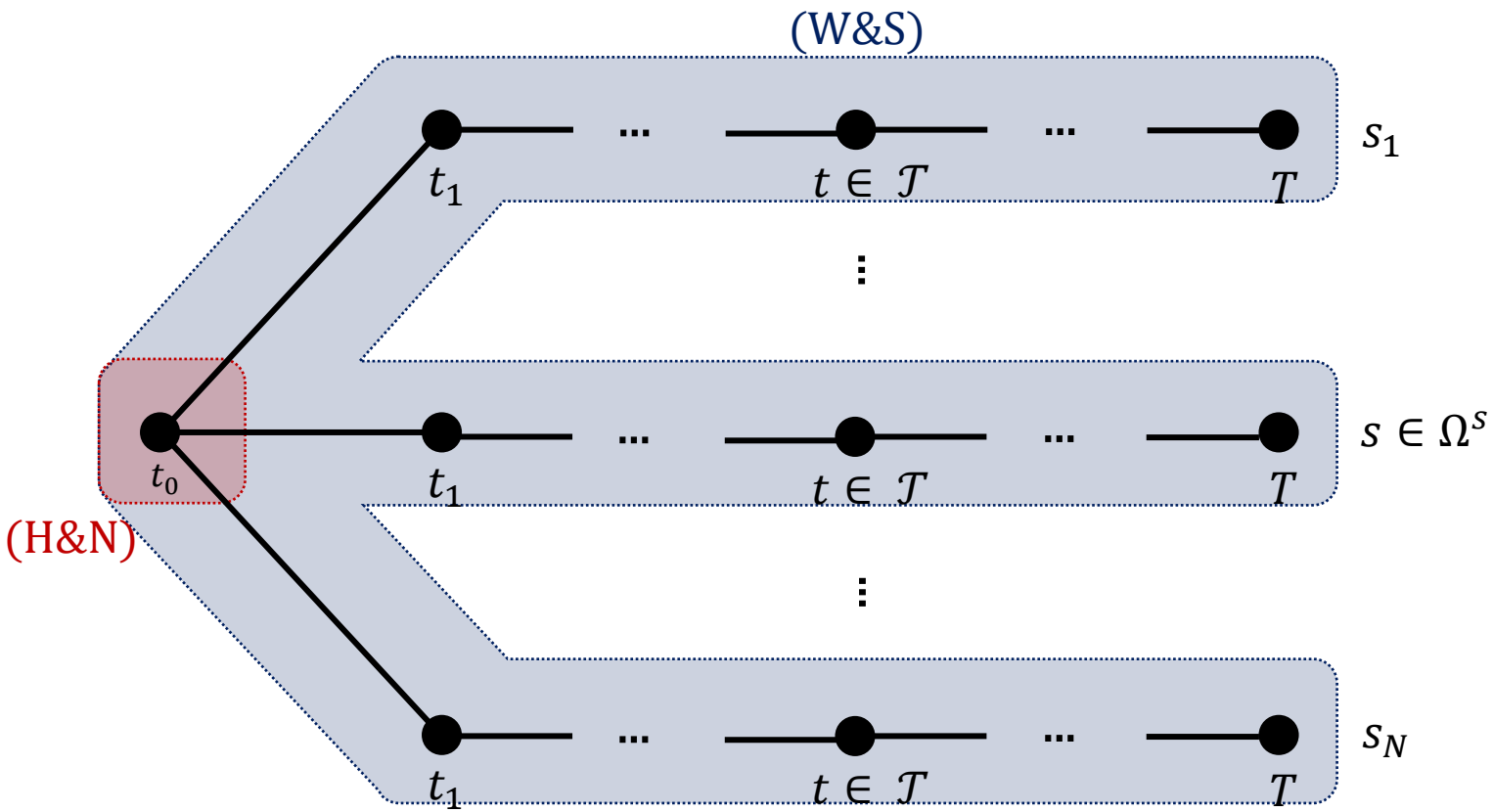
ADMM-based Stochastic scheduling

$$\min \sum_{s\in\Omega^s}\sum_{t\in\mathcal{T}} f(z_{t,s})$$

$$\text{s.t. } h(z_{t,s}) \le 0 \qquad t\in\mathcal{T}, s\in\Omega^s$$

$$x = z_{t0,s} : \lambda_s \qquad t\in\mathcal{T}$$

Consensus constraint



TU Delft

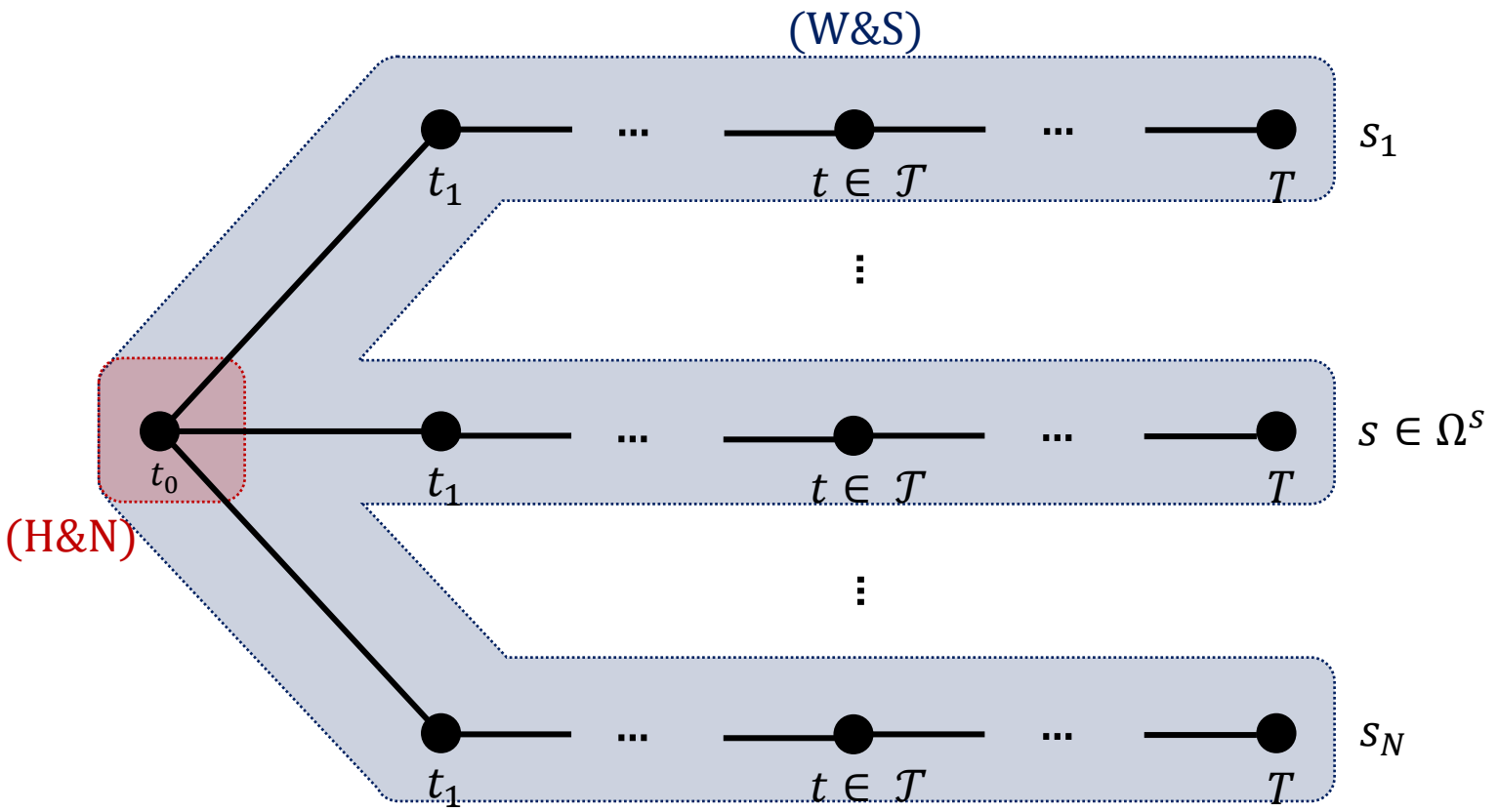
ADMM-based Stochastic scheduling

$$\min \mathcal{L}_\rho = \sum_{s\in\Omega^s}\left(\sum_{t\in\mathcal{T}} f(z_{t,s}) + \lambda_s(x - z_{t0,s}) + \frac{\rho}{2}(x - z_{t0,s})^2\right)$$

$$\text{s.t. } h(z_{t,s}) \le 0 \qquad t \in \mathcal{T}, s \in \Omega^s$$

$$\cancel{x = z_{t0,s} : \lambda_s \quad t \in \mathcal{T}}$$

~~Consensus constraint~~



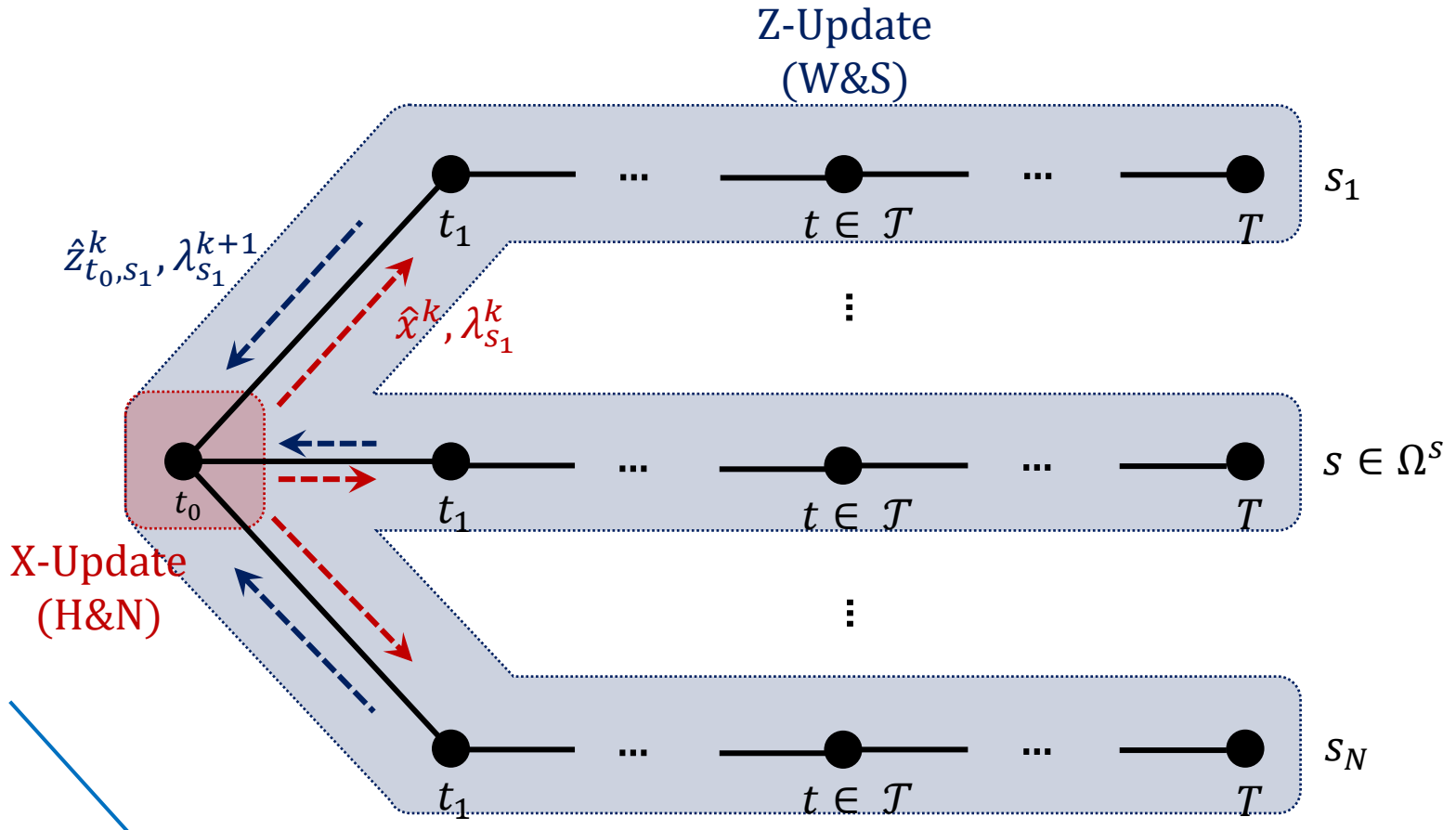
TU Delft

ADMM-based Stochastic scheduling

$$\min \mathcal{L}_\rho = \sum_{s\in\Omega^s}\left(\sum_{t\in\mathcal{T}} f(z_{t,s}) + \boldsymbol{\lambda_s}(x - z_{t0,s}) + \frac{\rho}{2}(x - z_{t0,s})^2\right)$$

$$\text{s.t. } h(z_{t,s}) \le 0 \qquad t\in\mathcal{T}, s\in\Omega^s$$

1. $x^{k+1} = \text{argmin}_x\, \mathcal{L}_\rho(x, z^k, \lambda^k)$
2. $z^{k+1} = \text{argmin}_x\, \mathcal{L}_\rho(x^k, z, \lambda^k) \quad s\in\Omega^s$
3. $\lambda^{k+1} = \lambda^k + \rho(x - z_s) \qquad s\in\Omega^s$



Each Z-update is MP-OPF.
While computationally expensive, needs less accuracy.

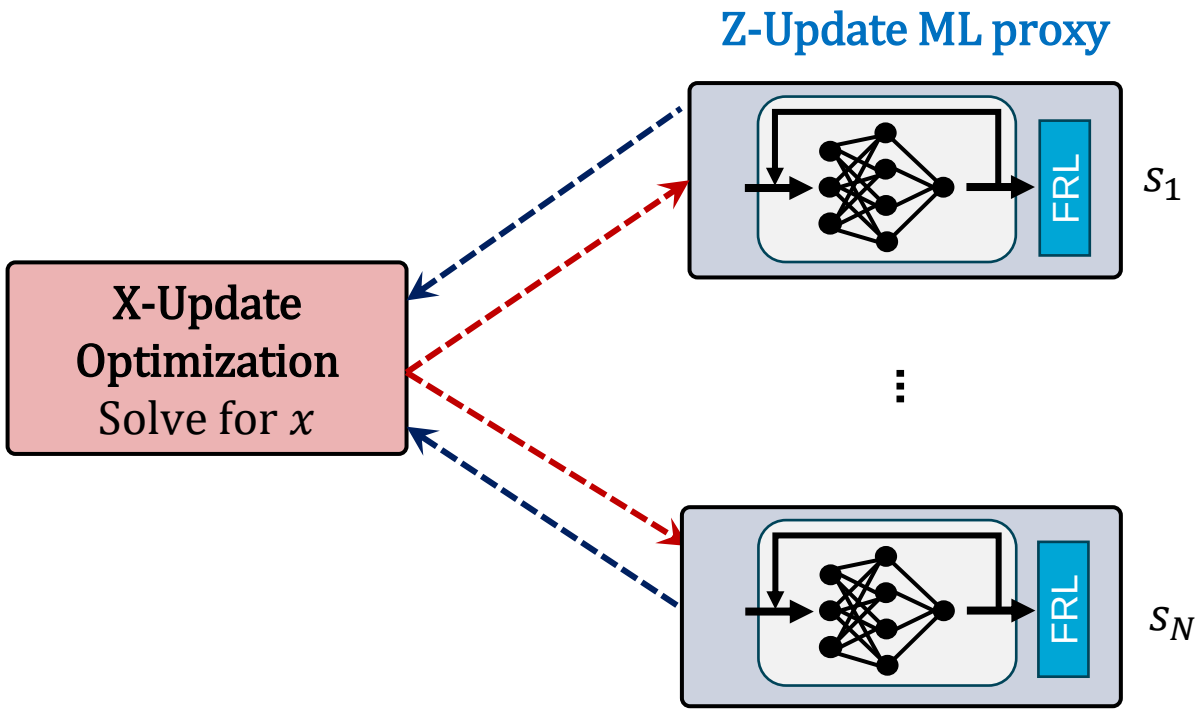
TU Delft

ADMM-ML

- H&N decisions by X-update optimization.
- Future W&S decisions by Z-update **ML proxy.**
  - Recurrent Neural Networks (RNN)

$$h_t = \sigma(Ax_t + Bh_{t-1})$$

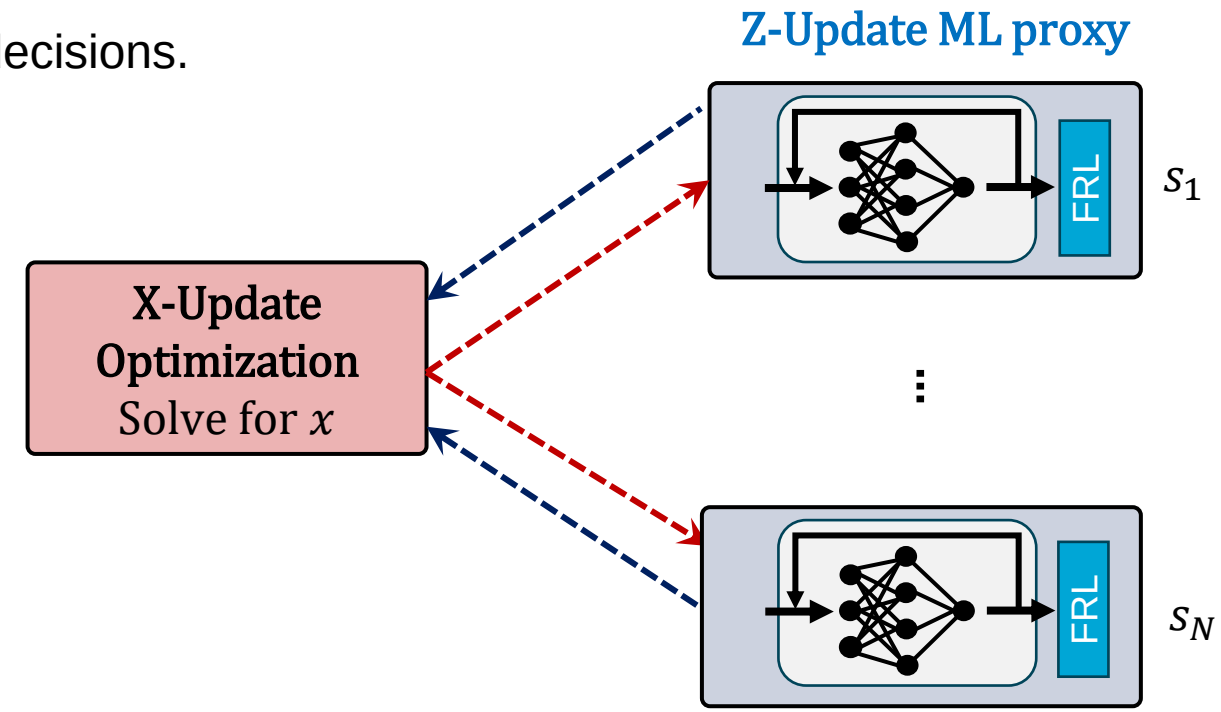
$$z_t = \sigma(Ch_t)$$



TU Delft

Properties of ADMM-ML

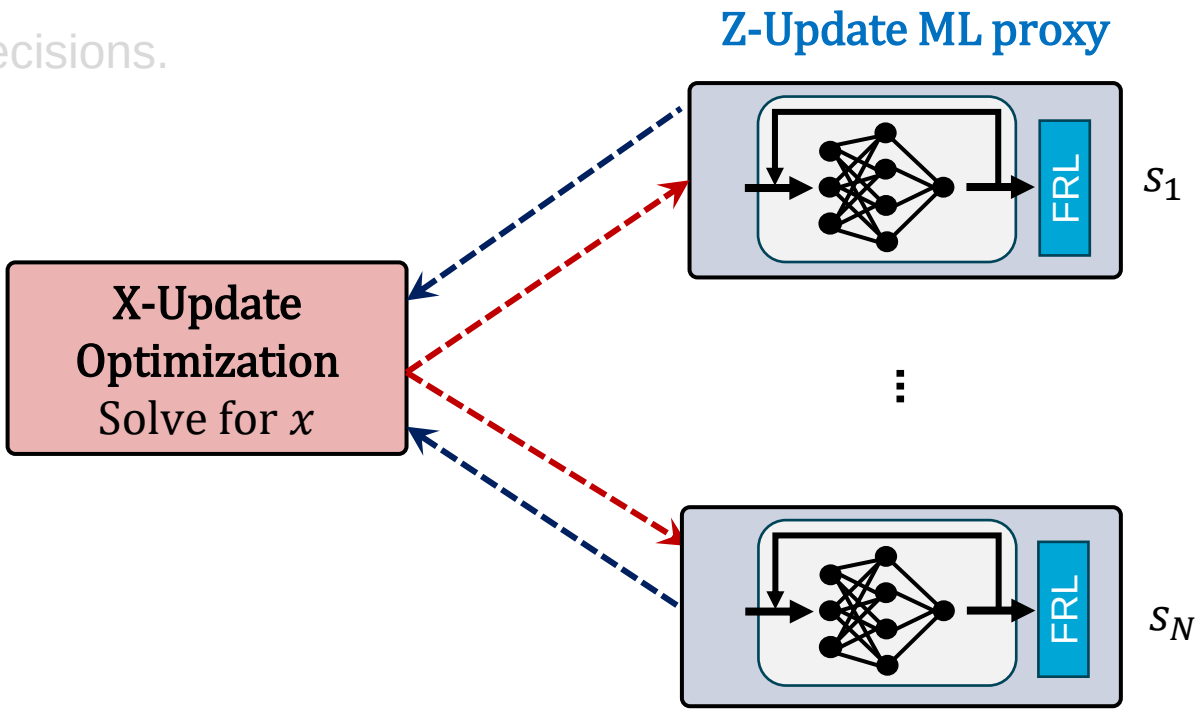
- **Physical Feasibility**
  - X-update optimization ensures feasibility for H&N decisions.



TU Delft

Properties of ADMM-ML

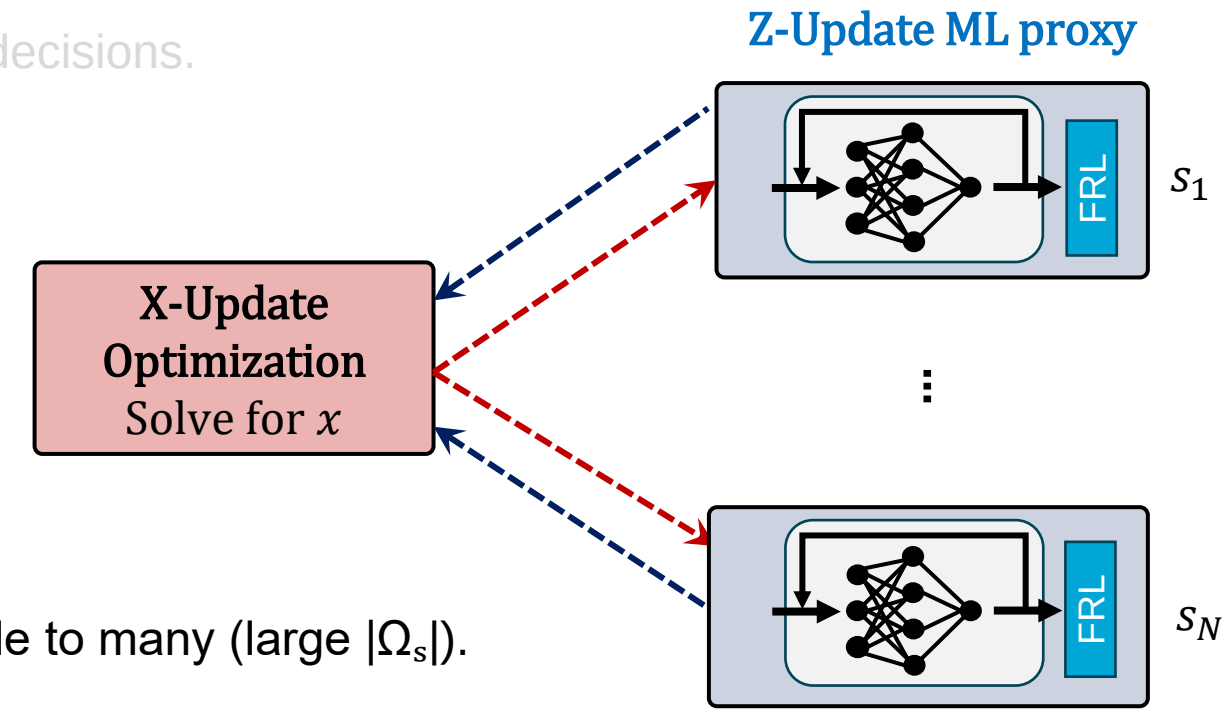
- Physical Feasibility
  - X-update optimization ensures feasibility for H&N decisions.
- **Fast & Parallelizable Inference**
  - RNN enables fast z-updates.
  - Scalable to long time horizons.
  - GPU parallelization.



TU Delft

Properties of ADMM-ML

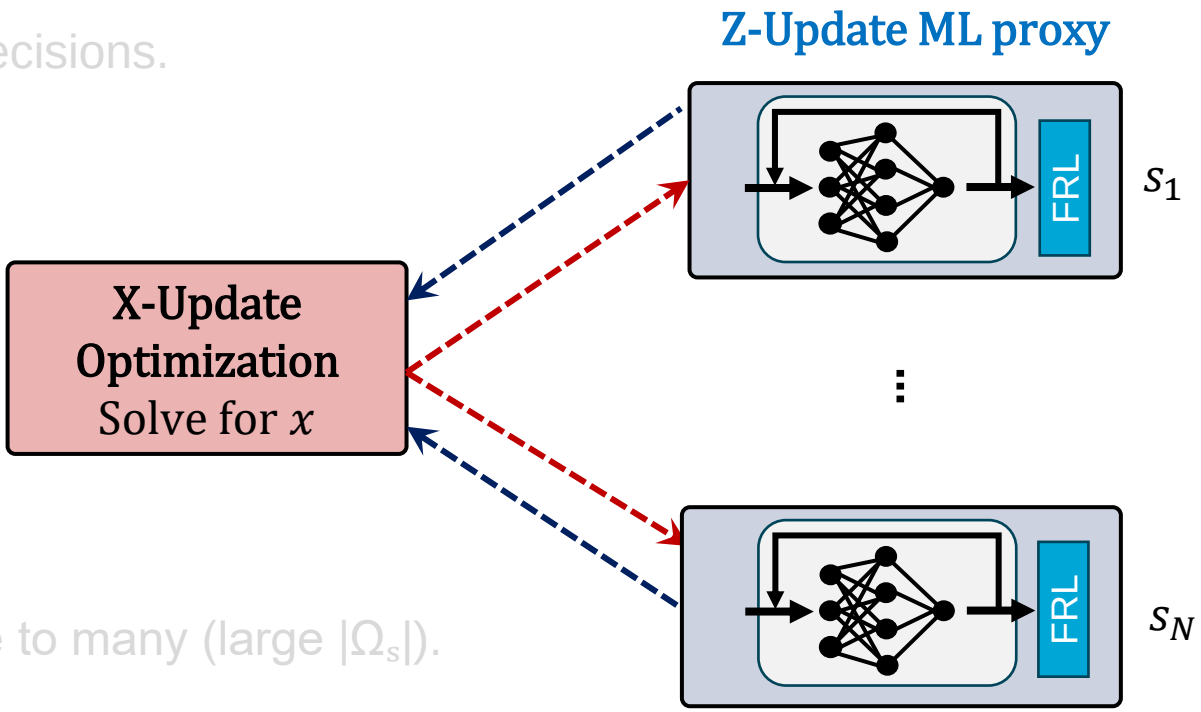
- Physical Feasibility
  - X-update optimization ensures feasibility for H&N decisions.
- Fast & Parallelizable Inference
  - RNN enables fast z-updates.
  - Scalable to long time horizons.
  - GPU parallelization.
- **Generalization Across Scenario Sizes**
  - Same RNN used across all uncertainty scenarios.
  - Trained on few scenarios (small $|\Omega_s|$), but applicable to many (large $|\Omega_s|$).



TU Delft

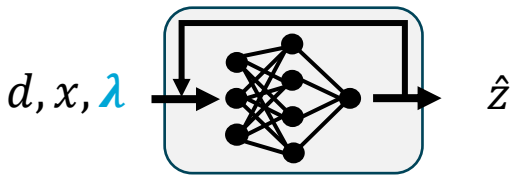
Properties of ADMM-ML

- Physical Feasibility
  - X-update optimization ensures feasibility for H&N decisions.
- Fast & Parallelizable Inference
  - RNN enables fast z-updates.
  - Scalable to long time horizons.
  - GPU parallelization.
- Generalization Across Scenario Sizes
  - Same RNN used across all uncertainty scenarios.
  - Trained on few scenarios (small $|\Omega_s|$), but applicable to many (large $|\Omega_s|$).
- **Warm Start for ADMM (Hybrid ADMM-ML).**



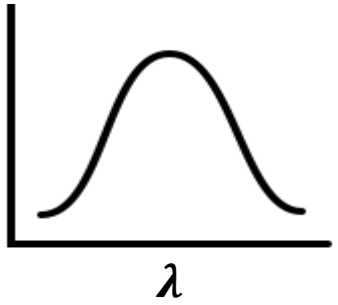
TU Delft

Training data generation



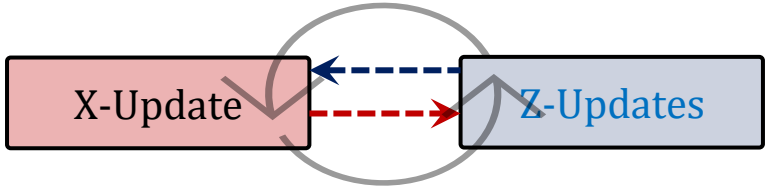
- Approaches:

1) Full exploration



ϵ-greedy exploration

2) Full exploitation

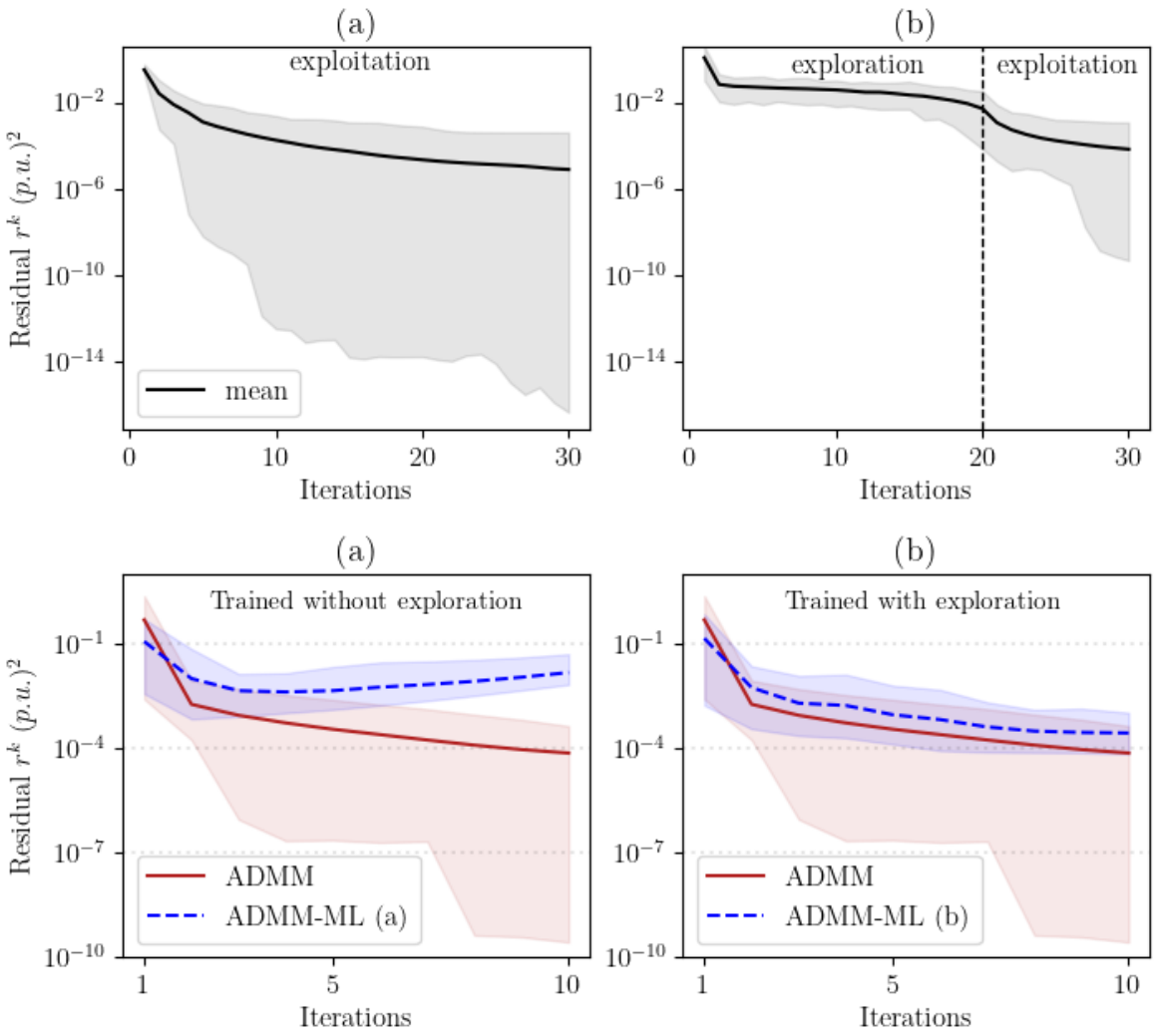


TU Delft

Case study: Training data generation

Training

Testing



TU Delft

Case study: Accuracy and time

- 118-bus St-MP-DCOPF.
- Test on samples with 500 scenarios.

CO
ADMM
ADMM-ML
ADMM-ML-H

TU Delft

Conclusion and future work

- Takeaway message: Combination of distributed optimization and ML may overcome the challenges of operational feasibility and scalability to numerous scenarios.

- Remaining challenges:
  - Practical applications: topology and system generalization.
  - Distribution shifts during online implementation.

TU Delft

A. Rajaei, O. Arowolo, and J. L. Cremer, "*Learning-accelerated ADMM for stochastic power system scheduling with numerous scenarios*," IEEE Transactions on Sustainable Energy, 2025. (Link)

Codes available on Delft-AI Lab GitHub. (Link)

Thank you for your attention!

Ali Rajaei
a.rajaei@tudelft.nl

TU Delft DAIE ENERGY

PowerTech KIEL 2025

https://www.tudelft.nl/ai/delft-ai-energy-lab